

Prior learning

SL & HL

Parallelogram's area	$A = bh$, b = base, h = height
Triangle's area	$A = \frac{1}{2}(bh)$, b = base, h = height
Trapezoid's area	$A = \frac{1}{2}(a + b)h$, a, b = parallel sides, h = height
Circle's area	$A = \pi r^2$, r = radius
Circle's circumference	$C = 2\pi r$, r = radius
Cuboid's volume	$V = lwh$, l = length, w = width h = height
Cylinder's volume	$V = \pi r^2 h$, r = radius, h = height
Prism's volume	$V = Ah$, A = cross-section area h = height
Cylinder curve's area	$A = 2\pi rh$, r = radius, h = height
Distance between 2 points (x_1, y_1) , (x_2, y_2)	$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
Coordinates for midpoints	$(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$

Topic 1: Number and algebra

SL & HL

Arithmetic sequence nth term	$u_n = u_1 + (n - 1)d$
Arithmetic sequence sum of n terms	$S_n = \frac{n}{2}(2u_1 + (n - 1)d) = \frac{n}{2}(u_1 + u_n)$
Geometric sequence nth term	$u_n = u_1 \cdot r^{n-1}$
Geometric sequence sum of n terms	$S_n = u_1 \cdot \frac{r^n - 1}{r - 1} = u_1 \cdot \frac{1 - r^n}{1 - r}, \quad r \neq 1$
Compound interest	$FV = PV \times \left(1 + \frac{r}{100k}\right)^{kn}$ FV = future value, PV = present value, n = # of years, k = # of compounding periods/year, r% = nominal annual rate of interest
Exponents and logarithms equivalence	$a^x = b \iff x = \log_a b, \quad a, b > 0, a \neq 1$
Logarithm properties	$\log_a(xy) = \log_a x + \log_a y$ $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$ $\log_a(x^m) = m \log_a x$ $\log_a x = \frac{\log_b x}{\log_b a}$
Infinite geometric series sum	$S_\infty = \frac{u_1}{1 - r}, \quad r < 1$
Binomial theorem	$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n$
Binomial coefficient	$\binom{n}{r} = {}_nC_r = \frac{n!}{r!(n - r)!}$

Topic 1: Number and algebra

HL

Combinations; Permutations	${}_nC_r = \frac{n!}{r!(n - r)!}; \quad {}_np_r = \frac{n!}{(n - r)!}$
Extension of Binomial Theorem, $n \in \mathbb{Q}$	$(a + b)^n = a^n \left(1 + n\left(\frac{b}{a}\right) + \frac{n(n-1)}{2!}\left(\frac{b}{a}\right)^2 + \dots\right)$
Complex numbers	$z = a + bi$
Modulus-argument form; Exponential form	$z = r(\cos \theta + i \sin \theta) = re^{i\theta} = r \operatorname{cis} \theta$
De Moivre's theorem	$[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta) = r^n e^{in\theta} = r^n \operatorname{cis} n\theta$

Topic 2: Functions

SL & HL

Equations of a straight line	$y = mx + c; \quad ax + by + d = 0$ $y - y_1 = m(x - x_1)$
Gradient formula	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Axis of symmetry of a quadratic function	$f(x) = ax^2 + bx + c \Rightarrow x = -\frac{b}{2a}$
Solutions of a quadratic equation	$ax^2 + bx + c = 0 : x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad a \neq 0$
Discriminant	$\Delta = b^2 - 4ac$
Exponential and logarithmic functions	$a^x = e^{x \ln a}; \quad \log_a a^x = x = a^{\log_a x}$ where $a, x > 0, a \neq 1$

Topic 2: Functions

HL

Sum and product of the roots of polynomial equations	$\sum_{r=0}^n a_r x^r = 0$ $\Rightarrow$ Sum is $-\frac{a_{n-1}}{a_n}$; product is $\frac{(-1)^n a_0}{a_n}$
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Topic 3: Geometry & trigonometry		SL & HL
Distance between two points $(x_1, y_1, z_1), (x_2, y_2, z_2)$	$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$	
Coordinates of midpoint $(x_1, y_1, z_1), (x_2, y_2, z_2)$	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$	
Volume: Right-pyramid	$V = \frac{1}{3}Ah, \quad A = \text{base area}, h = \text{height}$	
Volume: Right cone	$V = \frac{1}{3}\pi r^2 h, \quad r = \text{radius}, h = \text{height}$	
Area: Cone curve	$A = \pi r l, \quad r = \text{radius}, l = \text{slant height}$	
Volume: Sphere	$V = \frac{4}{3}\pi r^3, \quad r = \text{radius}$	
Surface area: Sphere	$A = 4\pi r^2, \quad r = \text{radius}$	
Sine rule	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$	
Cosine rule	$c^2 = a^2 + b^2 - 2ab \cos C$ $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$	
Area: Triangle	$A = \frac{1}{2}ab \sin C$	
Length of an arc	$l = r\theta, \quad r = \text{radius}, \theta = \text{angle in radians}$	
Area of a sector	$A = \frac{1}{2}r^2\theta, \quad r = \text{radius}, \theta = \text{angle in radians}$	
Identity for tanθ	$\tan \theta = \frac{\sin \theta}{\cos \theta}$	
Pythagorean identity	$\cos^2 \theta + \sin^2 \theta = 1$	
Double angle identities	$\sin 2\theta = 2 \sin \theta \cos \theta$ $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ $ = 2 \cos^2 \theta - 1$ $ = 1 - 2 \sin^2 \theta$	

Topic 3: Geometry & trigonometry		HL
Reciprocal trigonometric identities	$\sec \theta = \frac{1}{\cos \theta}; \quad \csc \theta = \frac{1}{\sin \theta}$	
Pythagorean identities	$1 + \tan^2 \theta = \sec^2 \theta; \quad 1 + \cot^2 \theta = \csc^2 \theta$	
Compound angle identities	$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$ $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$	
Double angle identity for tan	$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$	
Magnitude of a vector	$ \mathbf{v} = \sqrt{v_1^2 + v_2^2 + v_3^2}$	
Scalar product	$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3$ $\mathbf{v} \cdot \mathbf{w} = \mathbf{v} \mathbf{w} \cos \theta$ where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$	

Topic 3: Geometry & trigonometry		HL
Angle between two vectors	$\cos \theta = \frac{v_1 w_1 + v_2 w_2 + v_3 w_3}{ \mathbf{v} \mathbf{w} }$	
Vector equation of a line	$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$	
Parametric form of a line	$x = x_0 + \lambda l, \quad y = y_0 + \lambda m, \quad z = z_0 + \lambda n$	
Cartesian equations of a line	$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$	
Vector product	$\mathbf{v} \times \mathbf{w} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$ $ \mathbf{v} \times \mathbf{w} = \mathbf{v} \mathbf{w} \sin \theta$ where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$	
Area of a parallelogram	$A = \mathbf{v} \times \mathbf{w} , \quad \text{where } \mathbf{v} \text{ and } \mathbf{w} \text{ form two adjacent sides}$	
Vector equation of a plane	$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b} + \mu \mathbf{c}$	
Equation of a plane	$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} \quad (\text{using the normal vector})$	
Cartesian equation of a plane	$ax + by + cz = d$	

Topic 4: Statistics & probability		SL & HL
Interquartile range	$IQR = Q_3 - Q_1$	
Mean, $\bar{x}$, of a set of data	$\bar{x} = \frac{\sum_{i=1}^k f_i x_i}{n}, \quad \text{where } n = \sum_{i=1}^k f_i$	
Probability of an event A	$P(A) = \frac{n(A)}{n(U)}$	
Complementary events	$P(A) + P(A') = 1$	
Combined events	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	
Mutually exclusive events	$P(A \cup B) = P(A) + P(B)$	
Conditional probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$	
Independent events	$P(A \cap B) = P(A)P(B)$	
Expected value: Discrete random variable X	$E(X) = \sum xP(X = x)$	
Binomial distribution	$X \sim B(n, p)$ <i>Mean; Variance :</i> $E(X) = np; \text{Var}(X) = np(1 - p)$	
Standardized normal variable	$z = \frac{x - \mu}{\sigma}$	

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Topic 4: Statistics & probability		HL
Bayes' theorem	$P(B A) = \frac{P(B)P(A B)}{P(B)P(A B) + P(B')P(A B')}$ $P(B_i A) = \frac{P(B_i)P(A B_i)}{\sum P(B_j)P(A B_j)}$	
Variance σ^2	$\sigma^2 = \frac{\sum_{i=1}^k f_i(x_i - \mu)^2}{n} = \frac{\sum_{i=1}^k f_i x_i^2}{n} - \mu^2$	
Standard Deviation	$\sigma = \sqrt{\frac{\sum_{i=1}^k f_i(x_i - \mu)^2}{n}}$	
Linear transformation of a random variable	$E(aX + b) = aE(X) + b$ $Var(aX + b) = a^2 Var(X)$	
Expected value of continuous random variable X	$E(X) = \mu = \int_{-\infty}^{\infty} x f(x) dx$	
Variance	$Var(X) = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$	
Variance of discrete random variable X	$Var(X) = \sum (x - \mu)^2 P(X = x)$ $= \sum x^2 P(X = x) - \mu^2$	
Variance of continuous random variable X	$Var(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx =$ $\int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$	

Standard integrals	$\int \sin x dx = -\cos x + C$
	$\int \cos x dx = \sin x + C$
	$\int e^x dx = e^x + C$
Area enclosed by a curve and x-axis	$A = \int_a^b y dx$

Topic 5: Calculus
HL

Derivative of X from first principles	$\frac{dy}{dx} = f'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right)$
Standard derivatives	$f(x) = \tan x \Rightarrow f'(x) = \sec^2 x$ $f(x) = \sec x \Rightarrow f'(x) = \sec x \tan x$ $f(x) = \csc x \Rightarrow f'(x) = -\csc x \cot x$ $f(x) = \cot x \Rightarrow f'(x) = -\csc^2 x$ $f(x) = a^x \Rightarrow f'(x) = a^x (\ln a)$ $f(x) = \log_a x \Rightarrow f'(x) = \frac{1}{x \ln a}$ $f(x) = \arcsin x \Rightarrow f'(x) = \frac{1}{\sqrt{1-x^2}}$ $f(x) = \arccos x \Rightarrow f'(x) = -\frac{1}{\sqrt{1-x^2}}$ $f(x) = \arctan x \Rightarrow f'(x) = \frac{1}{1+x^2}$
Standard integrals	$\int a^x dx = \frac{1}{\ln a} a^x + C$ $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan \left(\frac{x}{a} \right) + C$ $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin \left(\frac{x}{a} \right) + C, \quad x < a$
Integration by parts	$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$
Area enclosed by a curve and y-axis	$A = \int_a^b x dy$
Volume of revolution about x or y axes	$V = \int_a^b \pi y^2 dx \quad \text{or} \quad V = \int_a^b \pi x^2 dy$
Euler's method	$y_{n+1} = y_n + h \times f(x_n, y_n); \quad x_{n+1} = x_n + h$ where h is a constant
Integrating factor for $y' + P(x)y = Q(x)$	$e^{\int P(x) dx}$
Maclaurin series	$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots$
Maclaurin series for special functions	$e^x = 1 + x + \frac{x^2}{2!} + \dots$ $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$ $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$

Topic 5: Calculus		SL & HL
Derivative of x^n	$f(x) = x^n \Rightarrow f'(x) = nx^{n-1}$	
Integral of x^n	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$	
Area between curve $y = f(x)$ and x-axis	$A = \int_a^b y dx, \quad \text{where } f(x) > 0$	
Derivative of $\sin x$	$f(x) = \sin x \Rightarrow f'(x) = \cos x$	
Derivative of $\cos x$	$f(x) = \cos x \Rightarrow f'(x) = -\sin x$	
Derivative of e^x	$f(x) = e^x \Rightarrow f'(x) = e^x$	
Derivative of $\ln x$	$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$	
Chain rule	$y = g(u), u = f(x) \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$	
Product rule	$y = uv \Rightarrow \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$	
Quotient rule	$y = \frac{u}{v} \Rightarrow \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$	
Acceleration	$a = \frac{dv}{dt} = \frac{d^2 s}{dt^2}$	
Distance; Displacement travelled from t_1 to t_2	$\text{dist} = \int_{t_1}^{t_2} v(t) dt; \quad \text{disp} = \int_{t_1}^{t_2} v(t) dt$	
Standard integrals	$\int \frac{1}{x} dx = \ln x + C$	